

Labwork 5

HPC – Gaussian blur convolution

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1 Preparation

The image is `astronaut()` from `skimage.data`, size 512×512 with 3 channels. The blur uses the 7×7 Gaussian kernel given in the labwork, whose weights add up to 1003, so after multiplying the window by the kernel I divide the sum by 1003 to keep the brightness.

Every output pixel is the weighted sum of the 7×7 window centred on it.

I wrote three versions: a CPU one as a reference, a GPU one reading from global memory, and a GPU one using shared memory. All three give exactly the same image (`np.array_equal` returns `True`).

2 Results

The CPU version takes 8971 ms.

The GPU is kept busy for one second first so that its clock is already up, then every configuration is launched once as a warm-up and 20 times for the measurement, and I keep the fastest one.

Table 1: Kernel time with and without shared memory (Tesla T4 on Google Colab).

Block size	Global memory (ms)	Shared memory (ms)	Speedup
(8, 8)	0.521	0.386	1.35x
(16, 8)	0.512	0.379	1.35x
(16, 16)	0.510	0.374	1.36x
(32, 8)	0.510	0.380	1.34x
(16, 32)	0.504	0.367	1.37x
(32, 32)	0.540	0.395	1.37x

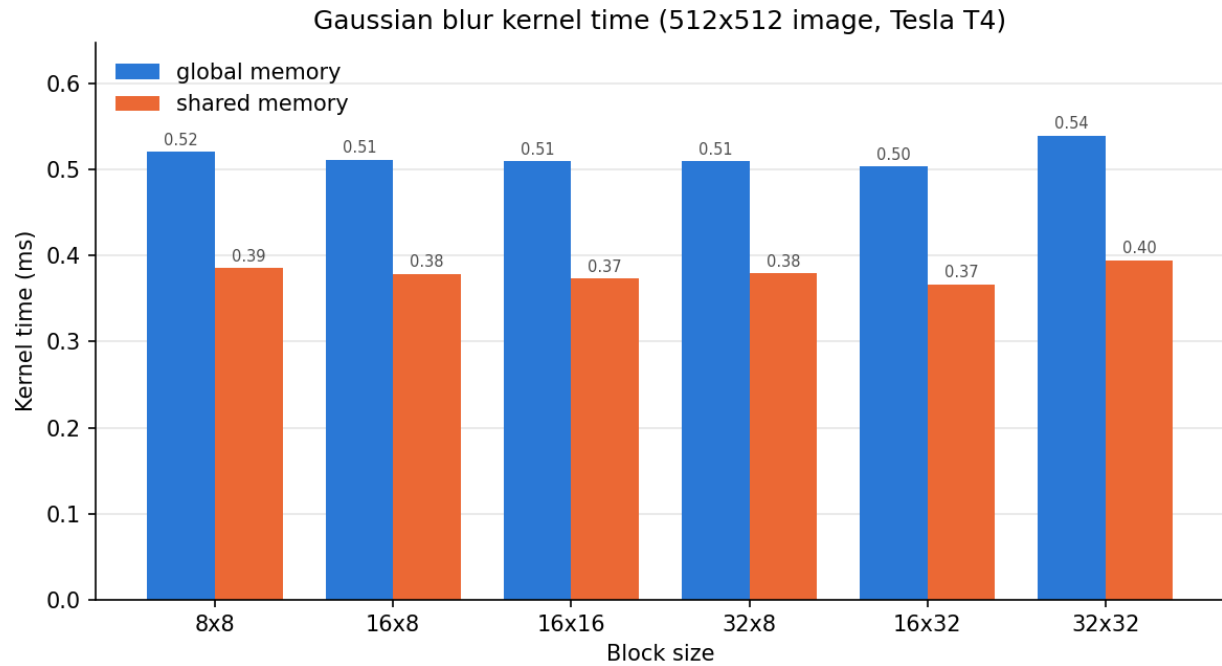


Figure 1: Kernel time without shared memory (blue) and with shared memory (orange).

The best configuration is (16, 32) for both versions: 0.504 ms without shared memory and 0.367 ms with it.

3 Comment

Shared memory makes the kernel about 1.35 times faster, and the gain is almost the same for every block size.